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Review

A foundational architecture for AI agents in healthcare

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SUMMARY

Medical AI agents represent a transformative paradigm in healthcare, distinguished from traditional AI by their autonomy, adaptability, and ability to manage complex tasks. This review introduces a conceptual framework for these agents built on four core components: planning, action, reflection, and memory. We examine the framework's application across key clinical domains, from enhancing diagnostic accuracy and personalizing treatment to guiding robotic surgery and enabling real-time patient monitoring. The review critically analyzes implementation challenges, including technical integration, clinician adoption, regulatory adaptation, and ethical considerations like data privacy and algorithmic bias. Future directions are explored, including the shift toward proactive, multi-agent collaborative systems and the visionary AI Agent Hospital concept. While these agents hold immense potential to revolutionize healthcare delivery by improving efficiency and patient outcomes, their successful and equitable integration hinges on navigating these profound technical, ethical, and regulatory hurdles.

INTRODUCTION

Artificial intelligence (AI) is transforming healthcare across numerous applications. The term "medical AI" broadly encompasses any use of AI in this domain, often referring to single-function algorithms designed for specific, isolated tasks—such as classifying medical images or predicting patient risk from a static dataset. These tools, while powerful, typically lack autonomy and contextual awareness. This review focuses on a more advanced and fundamentally different paradigm: the medical AI agent (Table 1). AI agents are intelligent systems designed to function autonomously or semi-autonomously, capable of making decisions and taking actions in dynamic, complex environments.[1–7] In the medical context, we define medical AI agents as autonomous computational systems structured around four core components: planning (cognitive processing and decision-making), action (execution through diverse interfaces), reflection (multimodal data perception and interpretation), and memory (contextual information storage and retrieval). Unlike traditional medical AI tools that operate on isolated datasets or

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Table 1. Comparison of human clinicians, traditional medical AI, and medical AI agents

Feature	Human clinician	Traditional medical AI	Medical AI agent
Core nature	biological intelligence with consciousness & empathy	a computational tool or algorithm	an autonomous or semi-autonomous system
Primary function	holistic patient care, complex reasoning, empathy	performs a specific, pre-defined task (e.g., classify, regression)	manages complex, multi-step tasks and workflows
Decision-making	experience-based, intuitive, and evidence-informed	rule-based or pattern recognition on a given dataset	data-driven, goal-oriented, and adaptive reasoning
Autonomy	full autonomy in clinical judgment	none; acts as a passive tool for human use	high; can operate independently to achieve goals
Learning & adaptation	lifelong learning from experience and education	static; requires retraining on new datasets to update	continuous learning from new data and interactions
Context awareness	high; understands patient history and context	low to none; typically context-agnostic	high; maintains context across interactions via memory
Interaction	direct interaction with patients and systems	human-invoked; typically a one-way information output	interactive; can query systems and collaborate with humans
Role in healthcare	the central decision-maker and caregiver	a supplementary tool to augment a specific task	a collaborative partner or assistant in the care process

perform single functions, medical AI agents integrate these four components to maintain context across interactions, learn from accumulated experiences, and adapt their behavior based on changing clinical scenarios, enabling comprehensive and dynamic healthcare support. Their core capabilities—autonomy, adaptability, and decision-making—allow them to perform tasks typically carried out by humans.[8–10] Autonomy enables agents to function independently, without the need for constant human supervision,[11] while adaptability ensures they can adjust their actions based on changing circumstances or new data.[12–14] Decision-making empowers agents to analyze data, evaluate various potential outcomes, and select the most effective course of action to achieve specific goals.[15,16] These abilities are powered by advanced technologies such as large language models (LLMs) and vision language models (VLMs), which enable agents to process and understand human language and visual information.[17–20] LLMs facilitate complex problem-solving, reasoning, and even creative thinking by interpreting and generating text, while VLMs extend these capabilities to visual data, enabling agents to make decisions based on images and videos.[20–23] Together, these technologies equip AI agents to perform tasks that require human-like intelligence and adaptability, forming the foundation for a wide range of applications.

The true potential of AI agents becomes evident when they operate across multiple modalities, integrating various forms of inputs such as text, images, speech, and sensory data.[24,25] This multimodal capability allows AI agents to process information in a way that mirrors human cognitive abilities, enabling more nuanced and effective decision-making.[26–29] For example, in robotics, AI agents can combine visual input, physical movement, and environmental feedback to execute complex tasks such as navigation, object manipulation, and autonomous control in dynamic environments.[30] Similarly, in the gaming industry, AI agents can seamlessly switch between understanding and responding to visual, auditory, and textual inputs, making real-time decisions that enhance the gaming experience by adapting to player behavior and environmental changes.[31] In autonomous vehicles, AI agents integrate data from cameras, LiDAR, and radar to navigate safely, avoiding obstacles and reacting to traffic conditions.[32–34] Furthermore, AI agents in smart homes use multimodal data to optimize user comfort and efficiency by integrating inputs from voice commands, sensors, and environmental data, ensuring automated systems function in harmony with human preferences.[35,36] This ability to synthesize diverse forms of data enables AI agents to tackle complex challenges across various fields, offering more robust, intelligent, and adaptable solutions.[37–39]

This review is based on a comprehensive literature search of PubMed, Google Scholar, and arXiv for articles published up to May 2025. Our search strategy utilized a combination of keywords related to the core concepts of medical AI agents, enabling technologies, key clinical applications, and implementation challenges. We included peer-reviewed articles, reviews, high-impact conference proceedings, and influential pre-prints relevant to AI agents in healthcare. The selected literature was then thematically synthesized to structure the primary sections of this review, covering the conceptual framework, clinical applications, development, evaluation, and ethical considerations of medical AI agents.

In the medical field, healthcare professionals are tasked with managing complex, time-sensitive situations that require accurate diagnoses, personalized treatment plans, and effective patient care[40–42] (Figure 1A). These challenges are amplified by the exponential growth of medical data and the increasing complexity of modern healthcare systems. Healthcare providers must process large volumes of diverse data, such as patient histories, medical imaging, laboratory results, and genetic information[43]—all of which must be integrated and analyzed to ensure optimal outcomes.[44,45] Furthermore, healthcare providers also need to adapt treatment plans in real time as patient conditions evolve.[44] Medical AI agents are uniquely positioned to address these multifaceted challenges through their advanced capabilities in data processing, autonomous decision-making, and adaptive learning. By leveraging multimodal technologies, AI

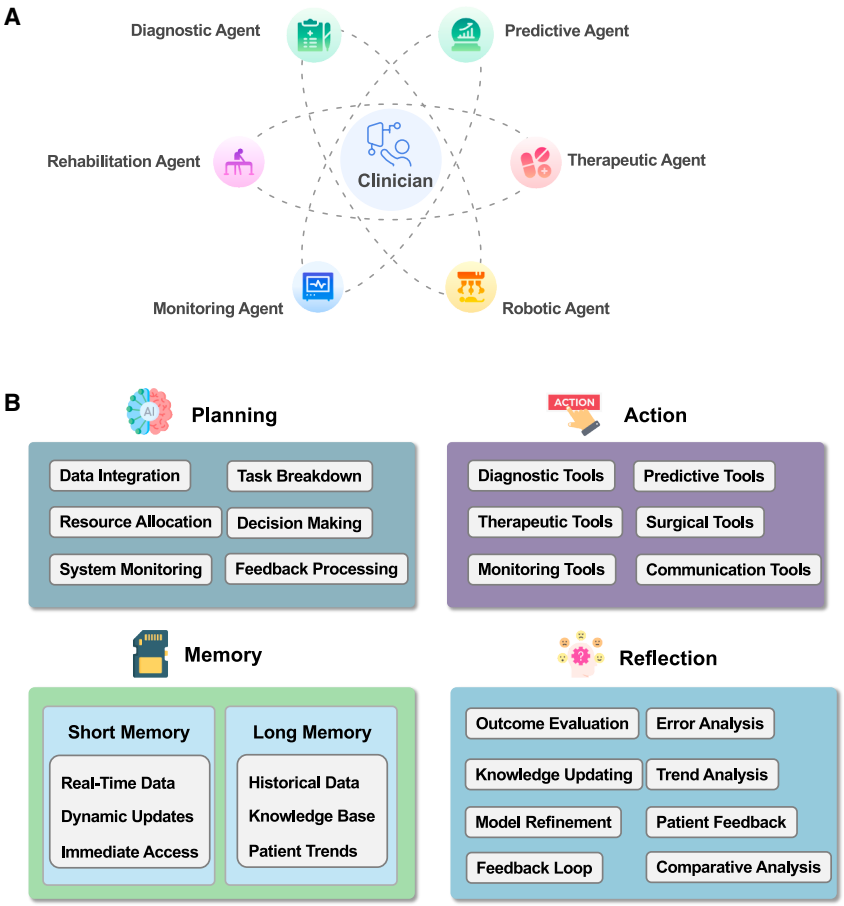


Figure 1. AI-driven multi-agent system in clinical practice

(A) This schematic illustrates an AI-driven multi-agent system that supports clinicians in patient care. The system includes diagnostic, predictive, therapeutic, monitor, rehabilitation, and robotic agents, each specializing in tasks such as disease identification, outcome prediction, personalized treatment, real-time monitoring, rehabilitation guidance, and surgical assistance. These agents work collaboratively, integrating multimodal data to enhance clinical decision-making, efficiency, and patient outcomes while keeping the clinician at the center of care.

(B) Medical AI agents operate based on four core components: planning, action, memory, and reflection. Planning enables strategic decision-making, resource allocation, and system monitoring. Action encompasses AI-driven tools for diagnosis, prediction, treatment, surgery, and patient monitoring. Memory is divided into short-term memory, which handles real-time data updates, and long-term memory, which stores historical data and patient trends for continuous learning. Reflection allows AI agents to evaluate outcomes, analyze errors, refine models, and incorporate patient feedback for improved decision-making. Together, these components ensure that AI agents function autonomously while continuously adapting to dynamic clinical environments.

agents can simultaneously analyze various data sources—such as medical imaging, electronic health records, and genetic data—to assist in diagnosis, predict patient outcomes, and recommend personalized treatment plans.[46–48] Their ability to autonomously make data-driven decisions and provide real-time insights enhances both the efficiency and accuracy of medical decision-making, supporting healthcare professionals in delivering timely, personalized care.[49] As these technologies continue to mature, they promise to revolutionize healthcare delivery by enabling more precise, efficient, and equitable patient care across diverse clinical settings.[50] This comprehensive review aims to establish a clear conceptual framework for medical AI agents, examine their current applications across healthcare domains, and identify future directions for their development and implementation in clinical practice. Our research question focuses on understanding how autonomous AI agents can be effectively integrated into healthcare delivery while addressing the technical, ethical, and regulatory challenges they present.

MEDICAL AI AGENT FRAMEWORK

AI agents are fundamentally structured around four core components—planning, action, reflection, and memory (Figure 1B)—that collectively enable their autonomous functionality in complex environments.[2,45,51–53] "Planning," often powered by LLMs or VLMs, serves as the cognitive core, allowing agents to process inputs, perform reasoning, and generate decisions. In healthcare, this capability enables agents to analyze medical records, synthesize information, and assist in diagnostic tasks with precision. "Action" translates these decisions into tangible outputs, such as generating responses, guiding robotic systems in surgical procedures, or recommending treatment plans. "Reflection" equips the agent with the ability to perceive and interpret its environment through sensory data, such as extracting insights from medical imaging or monitoring real-time patient conditions, facilitating context-aware interactions. Meanwhile, "memory" acts as a repository for past experiences and acquired knowledge, enabling agents to adapt and improve over time. This component is particularly critical in personalized medicine, as historical patient data are leveraged to refine recommendations and enhance outcomes. Together, these four components—cognition, perception, action, and memory—form an integrated framework, empowering AI agents to intelligently engage with their environment, adapt to dynamic challenges, and continuously evolve their capabilities.

Planning

The planning component of AI agents serves as the cognitive foundation, responsible for processing complex information, performing sophisticated reasoning, and making decisions.[5] At the core of this system are advanced models, particularly LLMs,[54,55] which enable agents to analyze complex inputs and draw meaningful inferences from extensive training data.[56,57]

This cognitive capability allows agents to handle diverse tasks, ranging from natural language understanding to specialized cognitive functions like logical reasoning or planning.[58] In healthcare contexts, the planning system processes patient data from electronic health records (EHRs), interprets diagnostic test results, and synthesizes information from medical literature to generate evidence-based recommendations. For example, when evaluating a patient with chest pain, the planning system can integrate symptoms, vital signs, laboratory results, and imaging findings to generate differential diagnoses and recommend appropriate diagnostic workups. The integration of external medical knowledge bases further enhances the agent's decision-making capabilities, ensuring recommendations align with current clinical guidelines and best practices.

Action

The action system of AI agents refers to their ability to execute decisions and perform tasks based on the data they process and the decisions they make.[38,39] In healthcare, this capability is exemplified through tools that allow the agent to carry out precise actions, from generating diagnostic reports to assisting in clinical procedures.[59] For instance, in robotic surgery, AI agents use their action systems to control robotic arms, executing delicate and highly precise movements based on real-time sensory feedback. Similarly, in clinical decision support, the action system can generate treatment recommendations or alert healthcare providers to critical changes in patient conditions. The tool component encompasses diverse interfaces including (1) application programming interfaces (APIs) for accessing electronic health records, laboratory information systems, and medical imaging repositories; (2) hardware interfaces for controlling medical devices such as infusion pumps, ventilators, and surgical robots; (3) specialized software libraries for image processing, natural language processing, and predictive analytics; and (4) robotic actuators for physical manipulation tasks in laboratory automation and patient care. These tools may include data retrieval systems, automated drug dispensers, or even integration with hospital management systems for streamlining administrative duties. By utilizing both internal action capabilities and external tools, AI agents can perform complex, multi-step procedures in healthcare environments, contributing to greater efficiency, accuracy, and improved patient outcomes.

Reflection

The reflection ability of AI agents is crucial for processing and interpreting sensory data in healthcare environments, enabling them to interact with and understand complex medical information.[15,60] This capability encompasses various modalities, including visual inputs, such as medical imaging (e.g., X-rays, magnetic resonance imaging [MRI], and computed tomography [CT] scans), and physiological data like heart rate, blood pressure, and other vital signs monitored through wearable devices. AI agents can analyze medical images to detect anomalies like tumors, fractures, or tissue damage, providing valuable insights for diagnosis and treatment planning. Additionally, the perceptual system allows the agent to process real-time data from sensors, enabling continuous monitoring of patient conditions and alerting healthcare providers to potential risks, such as deterioration in vital signs. By synthesizing diverse sensory inputs, AI agents can generate a more holistic understanding of a patient's health status, facilitating timely and accurate clinical decisions. This perceptual ability is vital for applications like robotic surgery, where the agent must interpret visual and tactile feedback to assist with precision in real-time operations.

Memory

The memory system of AI agents is fundamental for storing, retrieving, and applying knowledge over time, enabling them to adapt and improve based on past experiences.[4,8] The memory system operates through two distinct components: short-term and long-term memory. Short-term memory maintains immediate contextual information during active patient encounters, such as current vital signs (heart rate: 72 bpm, blood pressure: 120/80 mmHg), ongoing medication administration, and real-time symptom progression within a single clinical session. Long-term memory stores aggregated knowledge including patient medical histories, treatment outcomes across similar patient cohorts, learned patterns from thousands of diagnostic cases, and evidence-based treatment protocols that inform future decision-making. In healthcare, the memory system allows AI agents to retain patient data, treatment histories, and clinical outcomes, providing a valuable resource for informed decision-making. For example, an AI agent can recall a patient's medical history, previous treatments, and responses to medications, helping clinicians to tailor more personalized treatment plans.[61] Additionally, the memory system facilitates continuous learning, enabling the AI agent to refine its decision-making capabilities as new data become available. This learning is crucial for improving diagnostic accuracy, predicting patient outcomes, and identifying potential risks. By maintaining and updating a dynamic repository of knowledge, the memory system ensures that AI agents can leverage accumulated data for more precise and contextually appropriate actions. Thus, the memory system plays a pivotal role in enhancing the long-term effectiveness and reliability of AI agents in healthcare, ensuring they evolve in response to changing clinical needs and advancements in medical knowledge.

This four-component framework provides the theoretical architecture for these agents. The following section will demonstrate how this architecture is realized in practice across various clinical applications.

POTENTIAL APPLICATIONS OF MEDICAL AI AGENTS IN CLINICAL PRACTICE

As illustrated in Figure 2A, these individual applications do not operate in isolation but can be integrated into a cohesive, end-to-end patient care workflow orchestrated by a multi-agent system. This conceptual workflow traces a patient's journey from pre-admission, where an AI Early Warning Agent uses social and behavioral data to identify at-risk individuals, to hospitalization. Upon admission, a Diagnosis AI Agent synthesizes data from EHRs, lab tests, and imaging to perform differential diagnosis and risk stratification. This informs both the Treatment AI Agent, which devises personalized plans like targeted therapies, and the Surgical Robotics AI Agent, which assists in procedures

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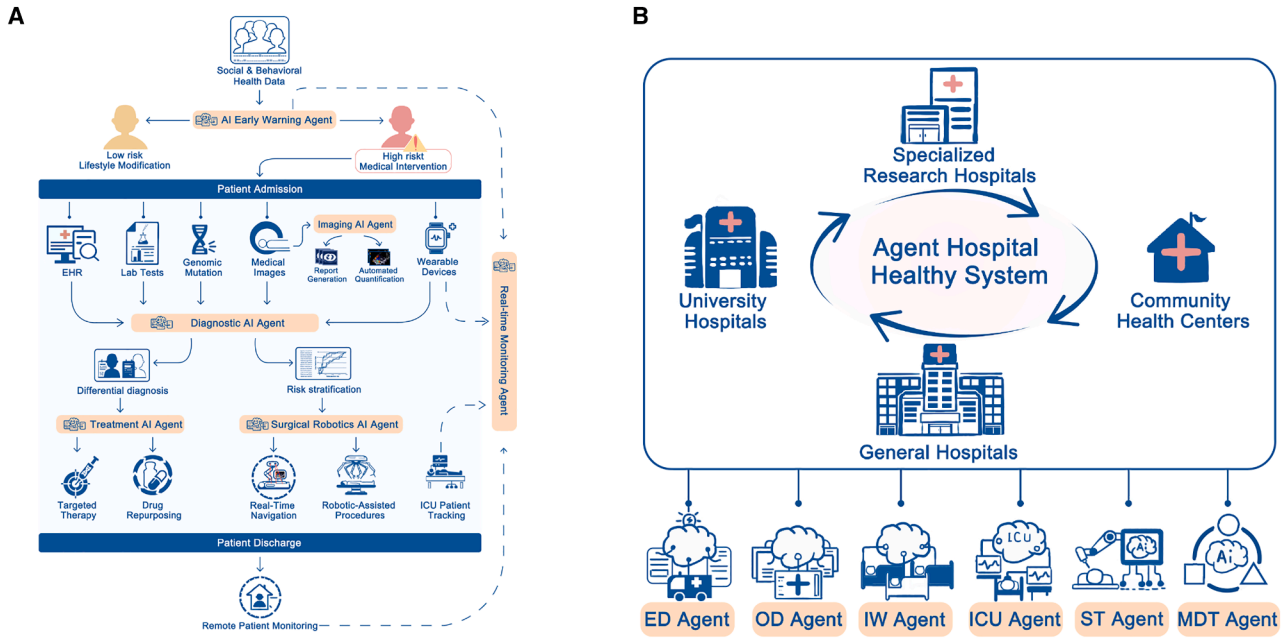


Figure 2. From a multi-agent clinical workflow to an AI Agent Hospital Ecosystem

This figure illustrates the transformative potential of medical AI agents at two interconnected scales.

(A) The patient-level clinical workflow: this panel depicts an end-to-end patient journey orchestrated by a team of specialized AI agents. It begins with pre-admission risk assessment by an AI Early Warning Agent and proceeds through in-hospital diagnosis, treatment, and surgery managed by dedicated agents, all under the continuous watch of a real-time monitoring agent. This demonstrates how multi-agent collaboration can create a seamless, personalized care continuum for an individual patient.

(B) The system-level AI Agent Hospital Ecosystem: this panel expands the concept to a system-wide level. The Agent Hospital Healthy System acts as a central coordinator, managing resources and patient flow across different healthcare facilities. At the operational level, function-specific agents such as the ED Agent, IW Agent, and MDT Agent handle the complex logistics of a hospital. (B) The ultimate vision where the patient-level workflows shown in (A) are scaled up and integrated across an entire health system, creating a truly intelligent, efficient, and interconnected healthcare network.

ED, emergency department; OD, outpatient department; IW, inpatient ward; ICU, intensive unit care; ST, surgical theater; MDT, multidisciplinary team.

with real-time navigation. Throughout the hospital stay and even after discharge, a Real-time Monitoring Agent continuously tracks patient status, ensuring timely interventions. This workflow-based model demonstrates how specialized AI agents can collaborate synergistically, transforming fragmented care into a continuous, proactive, and deeply personalized healthcare experience.

Diagnosis and decision support

As shown in Figure 2A, AI agents possess the capacity to process extensive patient data, including medical histories, diagnostic test results, and imaging, thereby aiding healthcare professionals in making more informed and precise diagnostic decisions.[62–69] For example, Google's Med-PaLM 2 demonstrated proficiency in medical question answering, achieving performance comparable to clinical experts on medical licensing examinations and performing better than generalist physicians in real-world medical questions.[70] In sepsis detection, AI agents like those deployed at Johns Hopkins Hospital continuously monitor patient vital signs and laboratory values, alerting clinicians to early sepsis indicators with promising sensitivity rates and potential for reducing mortality rates.[71] By synthesizing vast datasets drawn from medical literature, clinical guidelines, and real-time patient data, these agents significantly enhance diagnostic accuracy and efficiency.[72] Their ability to discern complex patterns, often unnoticed by humans, is particularly valuable in the diagnosis of rare or multifaceted conditions. Moreover, AI agents continuously evolve through iterative learning from new datasets, enabling them to adapt to emerging medical knowledge and evolving clinical standards.[73,74] This dynamic learning process ensures that AI agents remain aligned with the latest advancements in medical science, further enhancing their diagnostic capabilities.[75] By providing robust, evidence-based insights, AI agents help reduce diagnostic errors, enabling clinicians to deliver more timely and precise care. Through the integration of comprehensive data sources with real-time clinical information, AI agents serve as indispensable decision-support tools, enhancing both the accuracy and efficacy of the diagnostic workflow and ultimately improving patient outcomes.

Medical imaging analysis

AI-driven medical imaging analysis has emerged as a pivotal tool in modern healthcare, significantly improving the accuracy and efficiency of diagnostic processes.[66,74,76–83] By leveraging deep learning algorithms, AI systems can interpret complex medical images—such as X-rays, CT scans, MRIs, pathology images, and ultrasounds—providing automated, yet highly

precise assessments.[84,85] These systems are capable of identifying subtle patterns and anomalies that may elude human observation, enabling earlier detection of conditions such as tumors, fractures, and neurological disorders.[86] Additionally, AI algorithms improve continuously through exposure to diverse datasets, ensuring that their diagnostic capabilities evolve in response to new clinical evidence. By automating the image interpretation process, AI not only accelerates the diagnostic workflow but also reduces the potential for human error. Furthermore, AI's ability to integrate multimodal data, such as patient histories and genetic information, enhances the contextual understanding of imaging results, fostering more accurate and personalized treatment decisions. In this manner, AI is poised to redefine the landscape of medical imaging, enhancing both diagnostic precision and patient outcomes.[87]

Personalized treatment

AI agents show significant potential in the realm of personalized treatment by leveraging a variety of patient-specific data, including genetic information, medical history, lifestyle factors, and treatment responses.[77,88–92] With the ability to analyze and integrate this complex data, AI can assist in tailoring individualized treatment plans that optimize therapeutic outcomes. For instance, in oncology, AI agents can analyze a patient's genetic profile alongside clinical data to recommend targeted therapies that are more likely to be effective, minimizing adverse effects and improving prognosis.[93] Additionally, AI can predict how patients will respond to specific drugs based on their unique genetic makeup, facilitating the choice of the most suitable medication and dosage.[94] By continuously learning from patient data and treatment outcomes, AI agents help refine treatment plans in real-time, ensuring the most effective interventions are implemented throughout the course of care.[95] This ability to offer personalized, data-driven recommendations is transforming healthcare delivery, making it more precise and patient-centered.

Surgical assistance and robotic surgery

AI-driven surgical assistance and robotic surgery represent a paradigm shift in the precision and safety of medical interventions.[96–100] By integrating advanced algorithms with robotic systems, AI enhances the surgeon's ability to perform complex procedures with unparalleled accuracy.[101] These AI systems facilitate real-time image analysis, providing the surgeon with enhanced visualizations of the surgical site and critical anatomical structures, thus improving decision-making and reducing intraoperative risks.[102] Moreover, robotic surgery systems guided by AI can execute highly precise movements, minimizing human error and ensuring optimal outcomes, particularly in delicate, minimally invasive procedures.[103] The continuous learning capability of AI allows these systems to adapt based on historical surgical data, refining their performance over time. Furthermore, AI can predict potential complications during surgery by analyzing real-time data from patient monitoring systems, enabling proactive interventions. Through the integration of AI in surgical environments, healthcare delivery is transformed, leading to improved accuracy, reduced recovery times, and enhanced overall patient outcomes.

Real-time patient monitoring

AI-driven real-time patient monitoring systems are revolutionizing clinical care by enabling continuous, data-driven tracking of patient health.[90,104,105] Through integration of wearable devices, biosensors, and remote monitoring tools, AI agents can collect and analyze a range of physiological data, including heart rate, blood pressure, and blood glucose levels.[106] This real-time data analysis allows for early detection of health abnormalities, facilitating timely interventions and reducing risks.[107] AI algorithms can identify subtle patterns indicating the onset of critical events, such as arrhythmias or sepsis, which may otherwise go unnoticed. Moreover, through continuous learning, these systems refine their predictive capabilities, tailoring their assessments to individual patient conditions. By enhancing the accuracy of clinical decision-making, AI systems support healthcare providers in delivering more personalized, proactive care. Consequently, real-time patient monitoring powered by AI not only improves the responsiveness of healthcare teams but also contributes to better patient outcomes by enabling timely, evidence-based interventions.

AI Agent Hospital Healthy System

The applications discussed above—from diagnosis and treatment to surgery and monitoring—are not merely isolated advancements. This concept can be scaled to an entire healthcare system, culminating in the visionary model of the AI Hospital (Figure 2B). This idea is no longer purely theoretical; pioneering research, such as the "Agent Hospital" simulation by Li et al.[50] from Tsinghua University, has already demonstrated a virtual hospital where all roles—patients, nurses, and doctors—are played by autonomous LLM-powered agents. Inspired by such research, an overarching "Agent Hospital Healthy System" coordinates patient flow across a network of facilities, while function-specific agents—such as the ED (emergency department) agent, IW (inpatient ward) agent, and MDT (multidisciplinary team) agent—orchestrate complex daily logistics. The AI Agent Hospital, therefore, represents the ultimate application of medical AI agents: a fully integrated system where the micro-level efficiencies of individual patient care are aggregated to revolutionize healthcare delivery. This forward-looking application underscores the immense transformative potential of this technology, while also highlighting the scale of the challenges that will be discussed in the subsequent sections.

STAGES OF AUTONOMY IN MEDICAL AI AGENTS

The integration of AI into healthcare is marked by distinct levels of autonomy, each representing a deeper integration of AI agents into clinical workflows.[108,109] These levels range from the absence of AI intervention to full autonomy in patient care management. As AI systems evolve, they progressively take on more complex roles, supporting healthcare professionals in a wide range of tasks, from data analysis to decision-making.

In the initial stage, the use of AI is nonexistent, with healthcare professionals entirely responsible for diagnostic and treatment decisions. In this phase, there is no AI agent involved in the clinical process, and all tasks, including hypothesis generation, patient assessment, and treatment planning, are performed solely

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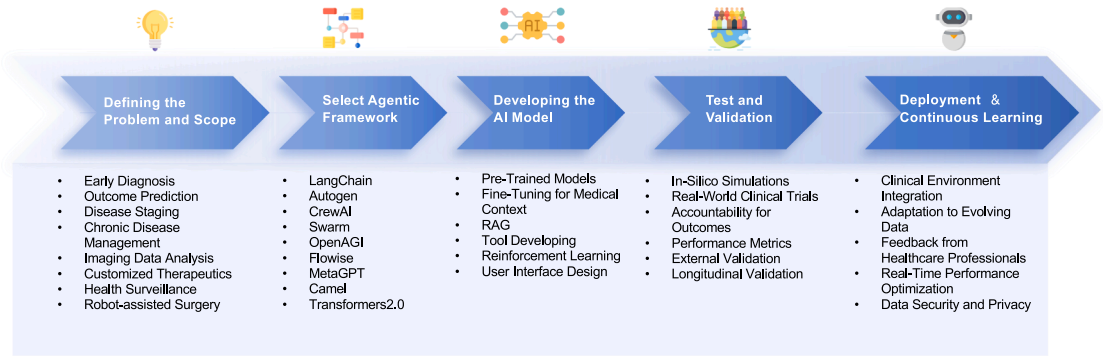


Figure 3. The development of medical AI agent

The development of medical AI agents follows a structured process comprising five key stages. Defining the problem and scope involves identifying clinical needs such as early diagnosis, outcome prediction, and robot-assisted surgery. Selecting an agentic framework entails choosing AI architectures like LangChain, AutoGen, and MetaGPT for efficient task execution. Developing the AI Model includes pre-training, fine-tuning for medical contexts, and reinforcement learning to enhance performance. Test and validation ensures safety and effectiveness through a structured pipeline that reflects the phased progression of clinical trials, moving from initial *in silico* simulations to controlled real-world clinical trials and subsequent external and longitudinal validation. Deployment and continuous learning integrates AI agents into clinical environments, allowing real-time optimization, adaptation to evolving data, and ensuring data security and privacy.

by human practitioners. AI has yet to play any role in augmenting clinical decision-making or workflow efficiency.

In the intermediate stage, AI agents act as assistants to healthcare providers. These systems assist with predefined tasks, such as analyzing patient data, identifying trends, and offering diagnostic suggestions based on established protocols. While AI provides valuable support, it remains a passive tool, with human clinicians making the final decisions and overseeing patient care.

In the advanced stage, AI agents transition into active collaborators, playing a significant role in clinical decision-making. At this stage, AI systems generate hypotheses, assist in designing treatment plans, and analyze complex datasets to provide real-time, data-driven insights. Healthcare providers and AI agents work in tandem, with AI continuously learning from new data to refine its recommendations, while human professionals retain control over final decisions and care pathways.

In the final stage, AI agents reach full autonomy, independently handling complex tasks such as diagnosing conditions, formulating treatment strategies, and adapting care plans in real time. These agents are capable of processing vast amounts of patient data, applying advanced machine learning algorithms, and autonomously adjusting patient care based on continuous monitoring and new information. Healthcare providers primarily focus on oversight and ethical validation, with AI executing the majority of clinical decisions, significantly enhancing both the efficiency and accuracy of healthcare delivery.

DEVELOPING MEDICAL AI AGENT: A COMPREHENSIVE APPROACH

The development of medical AI agents can be approached through two primary methodologies: graphical user interface (GUI)-based frameworks and coding-based development.[110–112] GUI-based approaches, commonly facilitated by low-code or no-code platforms, offer a streamlined pathway for constructing AI systems, allowing users to leverage intuitive drag-and-drop tools for rapid development. These platforms are particularly advantageous for prototyping and for those with limited programming expertise, facilitating quick deployment in specific medical use cases. However, this approach often sacrifices flexibility, scalability, and, importantly, data privacy, especially when integration with external services is required. In contrast, coding-based development provides a more robust and customizable solution, granting developers full control over the AI agent's architecture, decision-making processes, and data handling capabilities. This approach is particularly valuable in healthcare settings, where the security of sensitive patient data is paramount. Given its advantages in flexibility, control, and privacy, this part focuses on the coding-based development of medical AI agents, offering a comprehensive exploration of this methodology's capabilities and applications (Figure 3).

Defining the problem and scope

The initial phase in the development of a medical AI agent involves the precise identification of the specific healthcare problem that the agent is intended to address. This process extends beyond a narrow technical task; it requires contextualizing the agent's role within the broader ecosystems of digital health and clinical decision-making. In the context of digital health, defining the problem means understanding how the agent will function as a potential data integrator rather than another data silo. For example, is the agent designed to merely process a single data stream, or will it actively synthesize information from disparate sources like EHRs, wearable devices, and genomic databases to provide a holistic patient view? This consideration determines whether the agent supports a truly interconnected health model or simply adds to the existing fragmentation. Similarly, the scope must clarify if the agent's role is reactive (e.g., analyzing past events) or proactive (e.g., forecasting health risks), which aligns its development with the larger trend in digital health toward preventative care. Within the clinical decision-making process, defining the scope involves specifying the agent's level of partnership. Is it intended to be a passive information provider, merely flagging abnormalities for clinicians?

Or is it envisioned as a more active cognitive partner, capable of generating hypotheses, weighing evidence, and engaging in diagnostic reasoning alongside the healthcare team? This distinction is critical, as it directly impacts the design of human-agent interaction, the requirements for explainability, and the necessary evolution of clinical workflows to accommodate this new form of collaboration. Therefore, a well-defined problem statement that situates the agent within these broader contexts is crucial. It ensures the AI system is meticulously tailored to the complex, interconnected realities of modern healthcare, optimizing its utility and ensuring its relevance and efficacy.

Selecting the right agentic framework

After defining the problem, selecting the appropriate framework is crucial for developing a robust and effective medical AI agent. Agentic frameworks, such as LangChain and Microsoft AutoGen, provide the essential infrastructure required to construct AI systems capable of managing complex, dynamic workflows. These frameworks are designed with modular components, offering pre-built tools, memory management systems, and seamless data integration capabilities, thereby significantly reducing development time and simplifying the creation of intelligent agents. It is crucial that the chosen framework enables the AI agent to effectively interface with clinical databases, medical imaging tools, and real-time patient monitoring systems. Such integration is indispensable for generating actionable insights that can inform clinical decision-making and enhance the overall quality of patient care. The framework must not only support efficient data processing but also ensure that the AI agent can interact with diverse medical systems in a way that aligns with clinical workflows and regulatory standards.

Developing the AI model

Once the framework is established, the subsequent phase involves the development of the AI model that will underpin decision-making processes. This step requires the selection or training of an appropriate model based on the available medical data, including EHRs, patient histories, and diagnostic imaging.[113] The model may either be pre-trained—leveraging open-source or proprietary models—or constructed from the ground up using a dataset tailored to the specific clinical application. When using a pre-trained model, fine-tuning is often necessary to adapt the model to the particularities of the medical context and the defined task. Alternatively, training a bespoke model provides greater flexibility, enabling the AI system to learn directly from proprietary data, identify intricate patterns, and generate highly accurate predictions. Regardless of the approach, memory management is integral to the model's functionality, as the AI must retain and integrate contextual knowledge from past interactions and patient data. This capacity for contextual continuity is essential for delivering consistent, long-term care recommendations and ensuring the system's adaptability to the dynamic nature of patient conditions.

Test and validation

Prior to the deployment of a medical AI agent in clinical practice, it is imperative that the agent undergoes comprehensive testing and validation to ensure its safety, efficacy, and alignment with medical standards. This process involves both *in silico* simulations and clinical trials, allowing the AI to be evaluated in real-world conditions and ensuring that its performance meets the rigorous demands of healthcare environments. To ensure clinical relevance, this validation should ideally mirror the phased progression of traditional drug or device trials, beginning with initial feasibility and safety assessments before moving to larger-scale efficacy studies. The AI system must be meticulously validated against established medical protocols to confirm that its diagnostic predictions and therapeutic recommendations are consistent with current clinical expertise. Furthermore, adherence to regulatory frameworks, including compliance with data privacy laws such as HIPAA and medical device guidelines such as those set by the Food and Drug Administration (FDA), is essential for ensuring legal and ethical integrity. Just as importantly, ethical considerations such as transparency in decision-making, accountability for outcomes, and the mitigation of biases must be thoroughly addressed during this stage. This ensures that the AI operates in a manner that is scientifically sound and equitable in its application to patient care.

Deployment and continuous learning

Following successful validation, the medical AI agent is ready for deployment within a clinical environment. Seamless integration with existing healthcare infrastructure, including hospital management systems and EHRs, is crucial for efficient operation and data flow. Beyond technical integration, effective human-agent interaction design, grounded in principles from human-factors engineering,[114] is critical for successful clinical adoption. This requires intuitive interfaces that present AI recommendations in contextually appropriate formats with clear confidence indicators and alternative options, a design imperative focused on minimizing clinician cognitive load and enhancing situational awareness. Trust-building mechanisms must include gradual exposure protocols, performance transparency dashboards, and feedback loops where clinicians can correct AI decisions, enabling the system to learn from human expertise. Interface design should minimize cognitive load by presenting information hierarchically, highlighting critical alerts, and integrating seamlessly into existing electronic health record workflows, as supported by numerous usability studies in clinical settings.[115–117] Once deployed, continuous monitoring and regular feedback from healthcare professionals are essential to assess the AI system's ongoing effectiveness and relevance in real-world clinical settings. It is imperative that AI agents are designed to learn iteratively from new data and evolving clinical experiences, thereby enabling them to refine their capabilities and adjust to shifts in medical knowledge, practices, and patient demographics. This capacity for dynamic learning ensures that the AI system remains responsive and adaptive, optimizing its performance and clinical outcomes over time. Furthermore, as medical knowledge evolves and new challenges arise, the AI agent's continuous improvement contributes to more accurate decision-making and enhanced healthcare delivery, ultimately improving both patient care and operational efficiency.

EVALUATION OF MEDICAL AI AGENT

Recommendation accuracy

The fundamental criterion for evaluating a medical AI agent is its precision in generating diagnostic and treatment

recommendations.[80,100,118–120] This can be assessed by comparing the agent's outputs with those of experienced clinicians, particularly in the context of complex or rare conditions where human expertise is paramount. Diagnostic accuracy is typically quantified using established clinical metrics, including sensitivity, specificity, and overall accuracy. These measures provide a rigorous framework for assessing the AI system's performance, ensuring its recommendations align with current medical standards and evidence-based practices. Such evaluations are critical in confirming the AI's reliability as a clinical support tool, demonstrating its capability to offer valid, data-driven guidance in real-world medical settings.

Task execution and efficiency

The efficiency of a medical AI agent is not solely defined by the speed of task execution but also by the ability to reduce human workload, streamline medical workflows, and optimize overall operational efficiency.[121] The assessment of efficiency involves evaluating key performance metrics such as task completion time, success rate, step complexity, and economic cost. These factors are particularly relevant when AI agents rely on proprietary or closed-source models, which may introduce significant computational costs and licensing fees. For example, in applications such as automated medical record entry, drug prescription recommendations, and surgical planning, efficiency can be gauged by the time required to complete tasks, the number of steps needed, and the associated resource consumption. The economic cost, especially when leveraging closed-source models, must also be considered, as this can affect the cost-effectiveness of AI implementation. An efficient AI system that handles routine tasks with minimal time and resource expenditure enables healthcare professionals to focus more on patient care, thereby improving clinical outcomes and operational productivity.

Long-term learning and adaptation

Medical AI agents should be designed to continuously learn from new data and clinical experiences to sustain their relevance and effectiveness. However, evaluating long-term learning capabilities presents significant methodological challenges, requiring longitudinal studies with substantial costs and potential confounding variables from evolving medical practices. The evaluation of their capacity for ongoing improvement is based on the system's ability to incorporate model updates and integrate feedback loops.[122,123] AI systems that embed continuous learning processes are better equipped to adapt to emerging disease patterns, evolving treatment strategies, and changing patient outcomes. This adaptive capacity allows the AI to remain responsive to advancements in medical knowledge, ensuring its clinical applicability and enhancing patient care. Given the rapid pace of progress in medical tools and methodologies, as well as the variability of clinical contexts, it is essential that AI models are not static but modular. Such modularity facilitates the incorporation of specialized models tailored to distinct clinical scenarios, thereby increasing the agent's versatility and scope of application. This approach enables the AI agent to integrate new technologies and methodologies seamlessly, ensuring its sustained relevance across a wide array of healthcare applications, from diagnosis to treatment planning. To guide future research in this area, potential methodologies could include longitudinal simulation studies, which would allow for the assessment of an agent's adaptability in controlled, evolving virtual environments, thus overcoming the cost and complexity of long-term clinical trials.

Explainability and transparency

In the medical domain, the explainability of AI agents plays a crucial role in fostering their acceptance and ensuring their trustworthiness among healthcare professionals. AI systems are expected to provide transparent reasoning for their decisions, particularly when offering diagnostic suggestions or treatment strategies. The ability for clinicians to understand the rationale behind an AI's conclusions is essential for informed clinical decision-making, as it cultivates confidence in the system and supports appropriate human oversight. The evaluation of explainability involves assessing the AI's capacity to present its reasoning in a manner that is both accessible and comprehensible to medical practitioners. This may include the use of interpretable models and tools designed to visualize or explain the decision-making process, such as feature importance maps or decision trees. Explainability becomes even more critical in high-stakes environments, where AI-generated recommendations can significantly impact patient health. By ensuring that AI systems can provide clear, justifiable explanations, their clinical utility is enhanced, and the indispensable role of healthcare providers in decision-making is reinforced. Despite its importance, quantifying explainability remains inherently subjective, as different stakeholders—clinicians, patients, and regulators—may require fundamentally different types and levels of explanation. This subjectivity creates challenges in establishing standardized evaluation metrics that adequately capture the diverse needs and perspectives across healthcare settings. Future work could focus on developing stakeholder-specific explainability benchmarks, creating tailored evaluation frameworks that measure the utility and clarity of AI-generated explanations for distinct end-users, thereby moving beyond a one-size-fits-all approach to transparency.

While these applications highlight the immense potential of medical AI agents, their translation from promising prototypes to widespread clinical reality is fraught with significant hurdles. The following sections will critically analyze these technical, ethical, and practical challenges that must be addressed.

ETHICAL AND TECHNICAL CONSIDERATIONS

The implementation of AI agents in healthcare requires addressing various ethical and technical challenges to ensure they enhance healthcare delivery while maintaining trust, equity, and efficiency.[124] Real-world implementation faces substantial hurdles including IT infrastructure integration complexities, where legacy hospital systems often lack standardized APIs and interoperability standards. Cost-benefit analyses reveal significant upfront investments in computational infrastructure, staff training, and system maintenance, with implementation costs varying substantially depending on scope and complexity. Clinician adoption represents another critical barrier, as some

healthcare providers may express concerns about AI reliability and workflow disruption, necessitating comprehensive training programs and change management strategies to ensure successful integration.

Ethics

Medical AI agents must prioritize patient privacy and data security when they process sensitive information such as medical histories and treatment plans.[125–127] Robust encryption and strict access policies are crucial for preventing data breaches and misuse.[128] Transparency in decision-making is equally vital, requiring AI systems to provide clear explanations for their recommendations to ensure accountability and foster trust among patients and clinicians.[129] Another pressing concern is model bias, as AI systems may inadvertently reflect historical healthcare disparities, potentially leading to inequitable outcomes. Rigorous bias mitigation strategies are essential to ensure fairness across diverse patient populations. Additionally, the debate over closed-source versus open-source AI models presents unique ethical dilemmas. While closed-source systems may limit transparency, open-source models risk uncontrolled dissemination and misuse, necessitating a careful balance between innovation and security. Finally, concerns about AI replacing clinicians are unfounded, as AI agents are best positioned to augment, not replace, human expertise, allowing clinicians to focus on relational and complex aspects of care.

Technical challenges

The effective deployment of AI agents is hindered by several technical barriers. Efficiency remains a significant issue, as some tasks require extended processing times, impeding timely decision-making in critical medical scenarios.[130] Ensuring the quality of AI outputs is another challenge, as inconsistencies or ambiguities in results can have serious implications for patient care. Failure cases in medical AI systems underscore these challenges, including instances where AI diagnostic systems have demonstrated reduced accuracy across different patient populations and cases where AI-powered prediction algorithms have generated excessive false alarms, potentially causing alert fatigue among clinicians and affecting response to genuine emergencies. Limitations in context length also pose challenges, particularly for analyzing extensive clinical records and longitudinal data. Moreover, unexpected AI behaviors and deviations from instructions can disrupt workflows, leading to errors in diagnoses or treatment recommendations. Network reliability further complicates deployment, as latency or service interruptions can compromise real-time applications in healthcare. Addressing these challenges requires advances in model design, robust validation protocols, improved context management, and optimized network infrastructure.

Regulatory framework adaptation

Current medical device regulatory frameworks require substantial adaptation to accommodate the unique characteristics of AI agents with increasing autonomy and continuous learning capabilities. Traditional FDA pathways for medical device approval assume static algorithms with predictable behaviors, whereas autonomous AI agents evolve their decision-making patterns through continuous learning, presenting novel challenges for safety and efficacy monitoring. In response, regulatory agencies are establishing dedicated initiatives, such as the FDA's Digital Health Center of Excellence, to foster innovation while ensuring patient safety. These bodies are developing adaptive frameworks, but regional approaches differ. For instance, the FDA's Software as a Medical Device (SaMD) framework focuses on a risk-based and iterative life cycle approach, even allowing for pre-approved modification plans for adaptive AI algorithms. In contrast, the EU's AI Act provides a broader, cross-sectoral regulation that classifies AI systems into risk tiers, imposing stringent requirements on "high-risk" systems, which include many medical AI agents. While these frameworks establish risk-based classifications and require ongoing post-market surveillance, they must continue to evolve to fully address continuous learning algorithms that may develop unexpected behaviors, requiring dynamic validation protocols and real-time monitoring systems to ensure patient safety.

FUTURE DIRECTIONS FOR MEDICAL AI AGENTS

From reactive to proactive

By leveraging continuous monitoring of vital signs, environmental data, and patient histories, coupled with advanced predictive analytics, AI agents are capable of forecasting health risks and intervening before symptoms emerge. This paradigm empowers individuals to take control of their health and alleviates pressure on healthcare systems by prioritizing prevention over treatment. The transition to proactive medical AI represents a transformative step toward anticipatory, precision-focused healthcare that aligns with sustainability and patient-centric principles.

Personalized AI agents

Personalized medical agents, functioning as real-time health assistants, utilize advanced AI to monitor individual health metrics, including vital signs, activity levels, and environmental influences.[131] By analyzing data from wearable devices and biosensors, these agents deliver tailored health recommendations, predict potential risks, and issue timely alerts. This integration of continuous monitoring and predictive analytics enables individuals to proactively manage their well-being while facilitating early interventions, bridging the gap between clinical care and personal health autonomy for a more preventive, patient-centered approach.

Synergistic multi-agent collaboration

Collaborative multi-agent systems employ sophisticated coordination mechanisms including hierarchical task decomposition, distributed consensus algorithms, and dynamic role allocation protocols. The communication framework utilizes standardized medical ontologies to ensure semantic interoperability between agents. Conflict resolution mechanisms implement priority-based decision trees when agents provide contradictory recommendations, with escalation protocols involving human clinicians for complex cases. Diagnostic, therapeutic, and monitoring agents work in tandem, integrating their expertise to address complex medical challenges. For example, diagnostic agents

can analyze imaging data, while therapeutic agents develop tailored treatment plans, supported by real-time feedback from monitoring agents. This dynamic synergy enhances precision, efficiency, and adaptability across diverse clinical scenarios, from individualized care to large-scale public health interventions, paving the way for a more integrated and responsive healthcare ecosystem.

Lightweight models on edge devices

The deployment of lightweight AI models on edge devices revolutionizes medical AI by enabling real-time, localized data processing with minimal latency.[132,133] Optimized through techniques like pruning and quantization, these models deliver high performance while reducing computational demands, ensuring functionality even in low-resource settings.[103] By minimizing data transmission, they enhance patient privacy and mitigate connectivity limitations, making AI-powered diagnostics and personalized care accessible globally. Lightweight models on edge devices represent a transformative step toward equitable, efficient, and secure healthcare delivery.

Impact on medical education

The widespread adoption of AI agents necessitates fundamental changes to medical education curricula and training paradigms. Future clinicians must develop competencies in human-AI collaboration, including understanding AI limitations, interpreting algorithmic recommendations, and maintaining clinical reasoning skills despite AI assistance. Medical schools are beginning to integrate AI literacy courses covering machine learning principles, bias recognition, and ethical AI use. Additionally, training programs must emphasize skills that complement AI capabilities, such as empathetic patient communication, complex ethical decision-making, and adaptability to evolving technological landscapes. Simulation-based training with AI agents allows students to practice collaborative decision-making in controlled environments before clinical practice.

CONCLUSION

Medical AI agents are transforming healthcare by integrating advanced decision-making capabilities, autonomy, and adaptability into clinical workflows. Through multimodal technologies and a framework of planning, action, reflection, and memory, these agents address complex challenges such as personalized care, real-time monitoring, and predictive analytics. Our comprehensive review establishes a conceptual framework for medical AI agents, examines their current applications with specific case studies demonstrating promising clinical improvements, and addresses critical implementation challenges including IT integration complexities, clinician adoption barriers, and regulatory considerations. The evidence presented shows that medical AI agents can achieve significant clinical benefits in areas such as sepsis detection and emergency medicine response times, while also highlighting important limitations and failure cases that must be addressed. Future developments, including proactive systems, personalized agents, and lightweight edge models, emphasize a shift toward precision, efficiency, and equity in healthcare delivery. By fostering collaboration among diverse agents and leveraging cutting-edge technologies, AI agents hold immense potential to revolutionize patient care and improve health outcomes across a wide range of applications. However, successful implementation requires careful attention to human-agent interaction design, continuous learning evaluation, and adaptation of regulatory frameworks to accommodate the unique characteristics of autonomous AI systems in healthcare.

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AUTHOR CONTRIBUTIONS

Y.Y. and K.Z. conceived the idea. F.L. curated and analyzed the data, F.L. wrote the manuscript and created and generated the figures and tables. Y.N., Q.H.Z., J.L., K.W., Z.Y.D., I.N.W., L.L.C., T.L., L.D., K.L., G.L., T.W.H., M.F., H.Y.L., X.M.C., K.Z., and Y.Y. commented on and revised the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the authors used a large language model for grammar and language refinement in the writing process. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the final publication.

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